

MEHRAN RASOOL

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PROFILE

Computer Science undergraduate (5th semester) who builds and ships full, production-grade software independently, and applies the same rigor to coursework — turning class assignments into properly engineered, evaluated systems. Uses AI coding tools (Claude, Google Antigravity) as a core part of the workflow. Experienced across web backends, ML/NLP-based security systems, browser extensions, and game development.

INDEPENDENT PROJECTS

YT VISION — YouTube Analytics Platform

2026 — In Progress

Web app + browser extension + FastAPI backend | Python, FastAPI, React/Vite, Pydantic

- Designing a production-grade, multi-engine analytics architecture (SEO Engine, Engagement Engine, and more) inspired by vidIQ and TubeBuddy, built on SOLID principles with shared Pydantic contracts across a formal eight-stage pipeline; SEO and Engagement Engines are complete, independently tested, and integration-tested end-to-end against live YouTube data.
- Developed the 'Video Vision' React/Vite frontend dashboard, unifying SEO, Engagement, Velocity, and Thumbnail analysis into a single scrolling results view.

downsocial.net — Facebook Video Downloader

2026

Solo-built and deployed web service | Python, Flask, Nginx, Gunicorn, Cloudflare

- Built and deployed a production Flask backend with request rate limiting, an Nginx/Gunicorn server stack, and Cloudflare-based DMCA protection, with multilingual support for an international user base.

ACADEMIC PROJECTS

Secure LLM Gateway — Hybrid Multilingual AI Security Pipeline

4th Semester, 2026

Artificial Intelligence course project | Python, FastAPI, scikit-learn, Microsoft Presidio, Groq API

- Designed and built a five-stage hybrid security gateway for LLM applications, defending against prompt injection, jailbreaking, and PII leakage across multiple languages.
- Combined a deep-translator multilingual normalization layer, a TF-IDF + Logistic Regression semantic attack classifier, and Microsoft Presidio (with three custom recognizers for Pakistani CNIC numbers, API keys, and personal names) into a single scored decision pipeline, backed by a live Groq (Llama-3.1-8b-instant) API.
- Evaluated on a curated 150-prompt multilingual dataset, achieving 82.7% overall accuracy, 100% PII-masking recall, 100% block precision, and 525ms average latency; authored a full technical report on the architecture and evaluation.

Brute-Force Attack Detection System

4th Semester, 2026

Information Security course project | Built with Claude on Google Antigravity

- Built a web-based system to demonstrate, detect, and study brute-force login attack patterns as part of coursework on real-world attack vectors and defenses.

2D Dungeon Crawler Game

4th Semester, 2026

Data Structures & Algorithms course project | C++, SFML

- Built a multi-file, object-oriented 2D dungeon crawler applying core DSA concepts, with level-specific sprites, a particle system, an in-game shop UI, and level-unlock logic.

Real Estate Management System

2025

Object-Oriented Programming course project | Java

- Designed and built a Java OOP application for property listings, client records, and transaction tracking, applying core object-oriented design principles.

ATM Management System

2nd Semester

Programming Fundamentals course project | C++

- Built a console-based ATM simulation covering account management, balance inquiry, and transaction handling as a foundational programming project.

TECHNICAL SKILLS

Languages & Tools: Python, C++, Java, JavaScript, Flask, FastAPI, React, Vite, SFML, Pydantic, scikit-learn, Microsoft Presidio, Nginx, Gunicorn

AI-Assisted Development: Claude (Sonnet/Opus), Google Antigravity, Cursor Composer — used daily for architecture, coding, and debugging

Concepts: SOLID principles, REST API design, OOP, DSA, applied ML/NLP, LLM & prompt-injection security, web security basics

EDUCATION

BS Computer Science — COMSATS University Islamabad, Wah Campus

2024 — Present

5th Semester | Reg. No. SP24-BCS-008